MIL-S-19500/1A
4 May 1967
SUPERSEDING
MIL-T-19500/1
14 June 1957
(See 6.2)

MILITARY SPECIFICATION

# SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER

## TYPE 2N220

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for a low-power, PNP, germanium transistor.

1.2 Physical dimensions. See figure 1 (TO-1).

1.3 Maximum ratings.

| $P_C$ 1/ | $V_{CBO}$ | $V_{EBO}$ | $T_{stg}$ |
|---|---|---|---|
| mW | Vdc | Vdc | °C |
| 20 | -30 | -12 | -65 to +100 |

1/ Derate linearly 0.444 mW/° C for $T_A > 25°$ C.

1.4 Primary electrical characteristics.

| | $h_{FE}$ $V_{CE}$ = -4.0 Vdc $I_C$ = -0.5 mAdc | $C_{obo}$ $V_{CB}$ = -4 Vdc $I_E$ = 0.5 mAdc 1 kHz $\leq f \leq$ 1 MHz | NF $V_{CB}$ = -4 Vdc $I_E$ = 0.5 mAdc |
|---|---|---|---|
| | | pf | db |
| Min. | 40 | --- | --- |
| Max. | 120 | 50 | 6 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

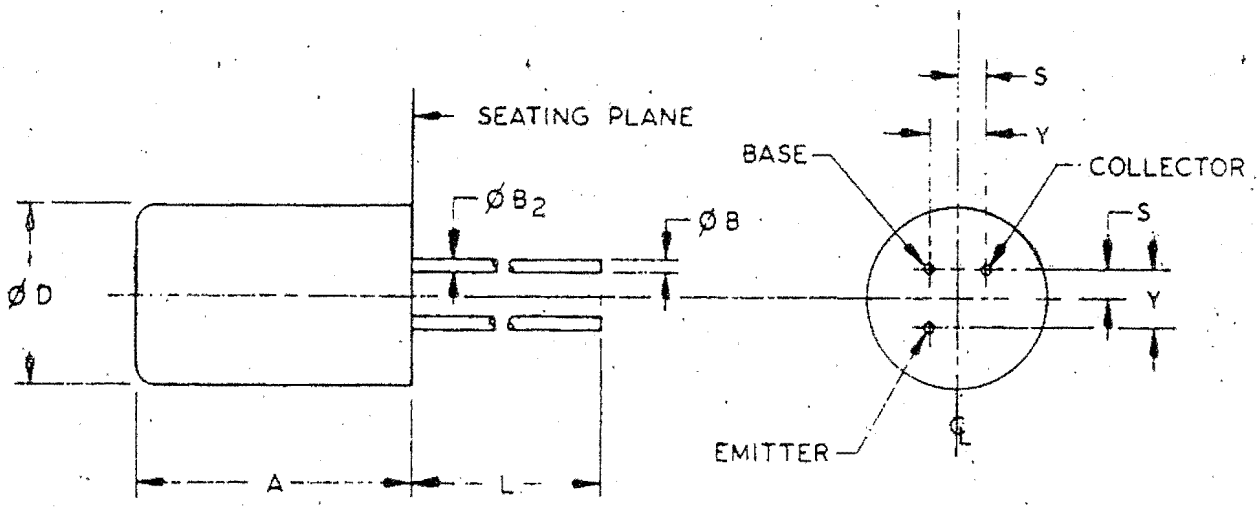
SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

FSC 5961

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| Ltr | Inches | | Millimeters | | Notes |
|---|---|---|---|---|---|
| | Min | Max | Min | Max | |
| A | -- | .410 | -- | 10.41 | |
| øB | -- | .021 | -- | .53 | 2,4 |
| øB2 | .016 | .019 | .41 | .48 | 3,4 |
| øD | -- | .240 | -- | 6.10 | |
| L | 1.500 | -- | 38.10 | -- | 4 |
| S | .0216 | .0286 | .55 | .73 | |
| Y | .0431 | .0573 | 1.09 | 1.46 | |

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. All three leads.
5. All leads shall be electrically isolated from the case.

FIGURE 1. Physical dimensions of transistor type 2N220 (TO-1).



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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II.



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TABLE I. Group A inspection

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 1 | | | 10 | | | | |
| Visual and mechanical examination | 2071 | | | --- | --- | --- | --- |
| Subgroup 2 | | | 5 | | | | |
| Collector-to-base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -12 | µAdc |
| Emitter-to-base cutoff current | 3061 | Bias cond. D; $V_{EB}$ = -12 Vdc | | $I_{EBO}$ | --- | -12 | µAdc |
| Subgroup 3 | | | 10 | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{fe}$ | 40 | 120 | --- |
| Open-circuit output capacitance | 3236 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc<br>100 kHz $\leq f \leq$ 1 MHz | | $C_{obo}$ | --- | 50 | pf |
| Noise figure | 3246 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc<br>$R_g$ = 1,000 ohms<br>$R_L$ = 20,000 ohms<br>f = 1 kHz | | NF | --- | 6 | db |
| Small-signal short-circuit input impedance | 3201 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{ib}$ | --- | 65 | ohms |
| Small-signal short-circuit output admittance | 3231 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{ob}$ | --- | 0.8 | µmhos |
| Small-signal open-circuit reverse voltage transfer ratio | 3211 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{rb}$ | 1.3 | 15 | $x10^{-4}$ |

TABLE II. Group B inspection

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 1 | | | 20 | | | | |
| Physical dimensions | 2066 | (See figure 1) | | --- | --- | --- | --- |



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TABLE II. Group B inspection - Continued

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 2 | | | 15 | | | | |
| Solderability | 2026 | Omit aging | | --- | --- | --- | --- |
| Thermal shock (temperature cycling) | 1051 | Test cond. B, except T(high) = 100 $^{+3}_{-0}$ °C | | --- | --- | --- | --- |
| Thermal shock (glass strain) | 1056 | Test cond. A | | --- | --- | --- | --- |
| Seal (leak-rate) | --- | Method 112 of MIL-STD-202; test cond. C, procedure III; test cond. B for gross leaks | | --- | --- | $5 \times 10^{-7}$ | atm cc/sec |
| Moisture resistance | 1021 | | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector-to-base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -12 | μAdc |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{fe}$ | 40 | 120 | --- |
| Noise figure | 3246 | $V_{CB}$ = -4 Vdc<br>$I_E$ = 0.5 mAdc<br>Rg = 1,000 ohms<br>$R_L$ = 20,000 ohms,<br>f = 1 kHz | | NF | --- | 6 | db |
| Subgroup 3 | | | 15 | | | | |
| Shock | 2016 | Nonoperating; 1,500 G approx. 0.5 msec, 5 blows in each orientation: $X_1$, $Y_1$, $Y_2$, and $Z_1$ | | --- | --- | --- | --- |
| Vibration fatigue | 2046 | Nonoperating | | --- | --- | --- | --- |
| Vibration, variable frequency | 2056 | | | --- | --- | --- | --- |
| Constant acceleration | 2006 | 20,000 G; in each orientation: $X_1$, $Y_1$, $Y_2$, and $Z_1$ | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 4 | | | 20 | | | | |
| Terminal strength (lead fatigue) | 2036 | Test cond. E | | --- | --- | --- | --- |



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TABLE II. Group B inspection - Continued

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 5 | | | 20 | | | | |
| Salt atmosphere (corrosion) | 1041 | | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 6 | | | $\lambda$ = 10 | | | | |
| High-temperature life (nonoperating) | 1031 | $T_{stg}$ = 100° C | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector-to-base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -14 | µAdc |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CB}$ = -4.0 Vdc<br>$I_E$ = 0.5 mAdc | | $h_{fe}$ | 32 | 140 | --- |
| Noise figure | 3246 | $V_{CB}$ = -4 Vdc<br>$I_E$ = 0.5 mAdc<br>Rg = 1,000 ohms<br>$R_L$ = 20,000 ohms<br>f = 1 kHz | | NF | --- | 7 | db |
| Subgroup 7 | | | $\lambda$ = 10 | | | | |
| Steady-state operation life | 1026 | $P_C$ = 20 mW min.<br>$V_{CB}$ = -12 Vdc | | --- | --- | --- | --- |
| End points: (Same as subgroup 6) | | | | | | | |

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Preparing activity:
Navy - SH

(Project 5961-0002-11)

Review activities:
Army - EL, MU, MI
Navy - SH
Air Force - 11, 17, 85

Code "C"

User activities:
Army - SM
Navy - CG, MC, AS, OS
Air Force - 14, 19



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